

# Two-Photon Physics at the $LH\mu C$

High-energy  $\gamma\gamma$  interactions in  $\mu p$  collisions

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(on behalf of HELP4CERN Study Group)

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# Why a muon-hadron collider for two-photon physics?



- A muon-hadron collider combines a **clean lepton-hadron initial state** with a multi-TeV centre-of-mass energy.
- Replacing the electron beam by a muon beam strongly relaxes the synchrotron-radiation limitation, making much higher lepton-beam energies possible.
- The  $\mu p$  environment remains experimentally cleaner than  $pp$  for many exclusive and semi-exclusive final states: negligible pile-up is a central advantage of the lepton-hadron concept.
- Both beams radiate photons. Their electromagnetic fields therefore create a high-energy  $\gamma\gamma$  collider *inside* the  $\mu p$  machine.

## Take-home message

LH $\mu$ C combines a high-energy photon source from the muon beam with the electromagnetic photon field of the proton, opening a distinctive high-energy  $\gamma\gamma$  programme.

Physics motivation adapted from the HELP4CERN / DIS2026 decks and the LHeC  $\gamma\gamma$  study.

# LH $\mu$ C@CERN: benchmark working points



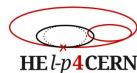
- **Workshop focus:**  $E_\mu = 500$  GeV and  $E_p = 7$  TeV,

$$\sqrt{s_{\mu p}} \simeq 3.7 \text{ TeV}, \quad \mathcal{L}_{\text{int}} \simeq 250 \text{ fb}^{-1}.$$

- Higher-energy HELP4CERN working points remain useful for scaling studies:
  - $E_\mu = 1 \text{ TeV} \Rightarrow \sqrt{s} \simeq 5.3 \text{ TeV}$ ,
  - $E_\mu = 2 \text{ TeV} \Rightarrow \sqrt{s} \simeq 7.5 \text{ TeV}$ .
- Already the 500 GeV option enters a qualitatively new  $\gamma\gamma$  mass regime compared with LHeC.

## Working points

|            | $E_\ell$       | $\sqrt{s}$     |
|------------|----------------|----------------|
| LHeC       | 50 GeV         | 1.2 TeV        |
| LH $\mu$ C | <b>500 GeV</b> | <b>3.7 TeV</b> |
| LH $\mu$ C | 1 TeV          | 5.3 TeV        |
| LH $\mu$ C | 2 TeV          | 7.5 TeV        |

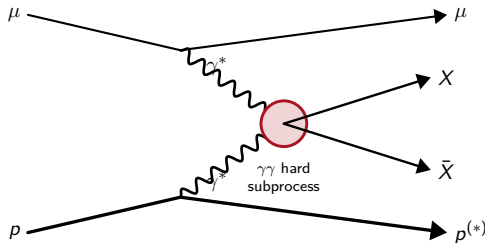


## This talk

The quantitative  $\gamma\gamma$  benchmarks shown below emphasise the 500 GeV  $\mu$  beam whenever dedicated results exist.

HELP4CERN benchmark set used in the DIS2026 HET study.

# A high-energy $\gamma\gamma$ collider inside $\mu p$ collisions



Both beams act as photon sources; the central system  
is produced by  $\gamma\gamma$  fusion.

## Factorised picture

$$\mu \rightarrow \mu + \gamma^*, \quad p \rightarrow p^{(*)} + \gamma^*,$$

$$\gamma\gamma \rightarrow X, \quad \mu p \rightarrow \mu + X + p^{(*)}.$$

- **Elastic:** proton remains intact.
- **Tagged elastic:** intact proton measured near the beam.
- **Dissociative:** proton breaks into a system of mass  $M_N$ .

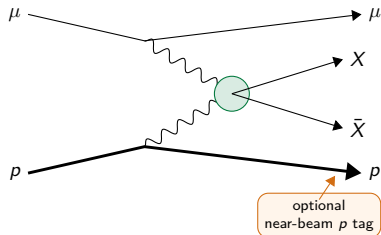
## Key point

One parent  $\mu p$  data set contains several distinct  $\gamma\gamma$  topologies with different photon virtualities and experimental handles.

# Two-photon production topologies: elastic, tagged and proton-dissociative

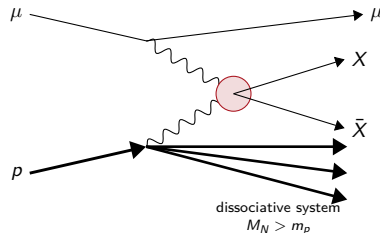


## Elastic / tagged elastic



- Proton electromagnetic form factors control the elastic flux.
- A near-beam tag can over-constrain event kinematics.

## Proton-dissociative



- Inelastic flux is described through proton structure functions.
- Larger  $M_N$  and  $Q_p^2$  increase the accessible luminosity but also modelling dependence.

Topology definitions follow the LHeC  $\gamma\gamma$  study; the same classification carries directly to  $\mu p$ .

# Equivalent Photon Approximation: factorising the $\mu p$ cross section

## EPA master relation

$$\sigma_{\mu p} = \int dy_\mu dy_p \Phi_\mu(y_\mu) \Phi_p(y_p) \sigma_{\gamma\gamma}(W) = \int dW S_{\gamma\gamma}(W) \sigma_{\gamma\gamma}(W)$$

- $\Phi_\mu$  and  $\Phi_p$  are the equivalent-photon fluxes radiated by the two beams.
- The hard subprocess  $\gamma\gamma \rightarrow X$  factorises from photon emission, in close analogy with partonic convolutions.
- $S_{\gamma\gamma}(W)$  packages the two fluxes into the effective two-photon luminosity spectrum.

EPA formalism: yy\_lhep / draft\_yy\_lhep.

## Kinematic variables

$$y_\mu = \frac{E_\gamma^{(\mu)}}{E_\mu}, \quad y_p = \frac{E_\gamma^{(p)}}{E_p},$$

$$W \equiv W_{\gamma\gamma}.$$

# Equivalent photon fluxes from the muon and proton

## Generic EPA flux

$$\Phi_\gamma(y) = \frac{\alpha}{\pi y} \int \frac{dQ^2}{Q^2} \left[ (1-y) \left( 1 - \frac{Q_{\min}^2}{Q^2} \right) F_E(Q^2) + \frac{y^2}{2} F_M(Q^2) \right]$$

$$Q_{\min}^2 = \frac{M^2 y^2}{1-y}.$$

- Point-like lepton:  $F_E = F_M = 1$  and  $M = m_\mu$ .
- The muon mass raises  $Q_{\min}^2$  relative to an electron at the same photon energy fraction  $y$ .

## Elastic proton emission

$$F_M = G_M^2,$$

$$F_E = \frac{4m_p^2 G_E^2 + Q^2 G_M^2}{4m_p^2 + Q^2}.$$

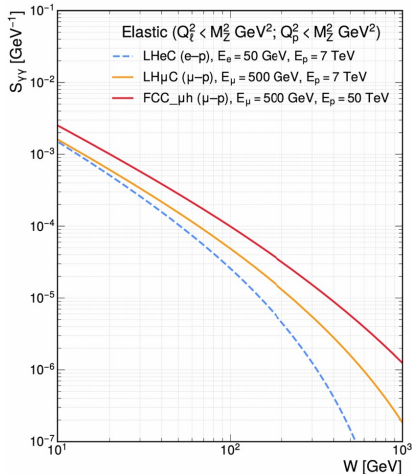
The dipole form factors suppress large  $Q_p^2$ .

## Proton dissociation

For  $p \rightarrow X$ , structure functions replace elastic form factors and the flux depends on the dissociative mass  $M_N$ .

Photon-flux formula and proton form factors: theory input of yy\_lhec / draft\_yy\_lhec.

# The $\gamma\gamma$ luminosity spectrum $S_{\gamma\gamma}(W)$



## $S_{\gamma\gamma}$ tells us

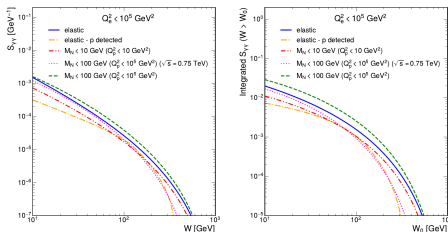
The fraction of the parent lepton-proton luminosity available for two-photon collisions at invariant mass  $W$ .

- The 500 GeV LH $\mu$ C spectrum extends well beyond the LHeC reach.
- A harder spectrum benefits the  $WW$ ,  $ZZ$ ,  $t\bar{t}$  and charged-BSM thresholds.
- Energy-growing EFT effects are naturally amplified in the high- $W$  tail.

**LH $\mu$ C message:** the gain is not only in total statistics; the  $\gamma\gamma$  spectrum is substantially harder.

Existing HELP4CERN elastic-luminosity comparison; orange:  $E_\mu = 500$  GeV,  $E_p = 7$  TeV.

# Elastic vs. inelastic $\gamma\gamma$ luminosity and forward proton tagging



## LHeC reference study

The detailed calculation separates elastic, tagged elastic ( $0.01 < y_p < 0.1$ ), and proton-dissociative photon emission for several  $M_N$  and  $Q_p^2$  ranges.

- Larger allowed dissociative masses increase the usable  $\gamma\gamma$  luminosity.
- A forward proton tag improves kinematic closure, especially for final states with neutrinos.
- Low pile-up means proton tagging need not be mandatory for every clean channel.

LHeC  $\gamma\gamma$  paper: luminosity spectrum and near-beam tagging discussion; topology logic transfers to LH $\mu$ C.

# Beyond the simplest EPA: photon virtuality and exact $W_{\gamma\gamma}$ kinematics

## High-energy approximation

$$W^2 \simeq y_\ell y_p s.$$

## Exact elastic relation in the updated study

$$W^2 = -Q_\ell^2 + y_\ell y_p s$$

The paper writes  $\ell = e$ ; the same lepton-side kinematics applies after  $e \rightarrow \mu$ .

Updated exact-kinematics study: draft\_yy\_lhec, Appendices C–E.

- The approximation is excellent when photon virtualities are negligible relative to  $W^2$ .
- For proton dissociation, the exact expression also depends on  $Q_p^2$  and  $M_N$ .
- Elastic EPA agrees at percent level with matrix-element generators once the proper  $W$  definition is used; inelastic modelling is more sensitive to the dissociative phase space.

## Precision lesson

At sizeable  $Q^2$ , correct  $W_{\gamma\gamma}$  reconstruction is part of the physics model, not a cosmetic refinement.

# From photon luminosity to physics: $\gamma\gamma \rightarrow X$



## Collider rate

$$\sigma(\mu p \rightarrow \mu X p^{(*)}) = S_{\gamma\gamma} \otimes \sigma(\gamma\gamma \rightarrow X).$$

$\ell^+\ell^-$   
calibration

$\tau^+\tau^-$   
dipole  
moments

$W^+W^-$   
gauge  
structure

$ZZ$   
loop / nQGC

$t\bar{t}$   
heavy  
threshold

charged BSM  
pairs

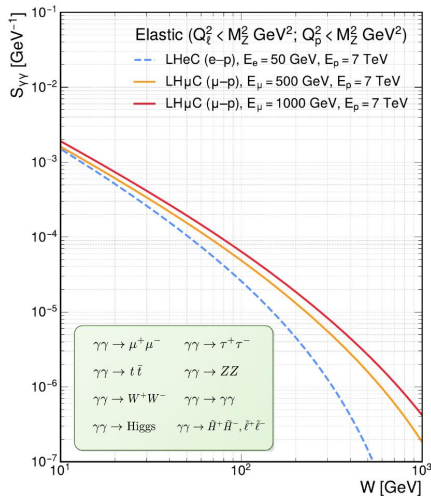
$\gamma\gamma$   
light-by-light

$H$   
loop-induced

## Take-home message

The same photon luminosity supports precision QED, electroweak and loop physics, heavy thresholds, and direct searches for charged new states.

# The high-energy $\gamma\gamma$ physics landscape at LH $\mu$ C



Existing HELP4CERN threshold summary, adapted to the dedicated two-photon narrative.

- $\ell\ell, \tau\tau$ : high-rate QED calibration and dipole probes.
- $WW$ : sizeable SM rate and direct  $\gamma\gamma WW$  coupling.
- $ZZ$ : loop-induced SM benchmark and neutral quartic couplings.
- $t\bar{t}$ : heavy threshold and top electromagnetic structure.
- **Charged BSM pairs**: production controlled mainly by electric charge and mass.
- $\gamma\gamma$  and  $H$ : loop-induced quantum processes and complementary searches.

# $\gamma\gamma \rightarrow \ell^+ \ell^-$ : a standard candle for two-photon physics



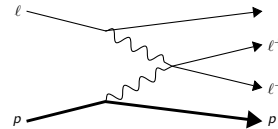
- $e^+e^-$  and  $\mu^+\mu^-$  provide clean control samples for the complete  $\gamma\gamma$  programme.
- They test EPA modelling, acoplanarity and elastic/inelastic separation.
- At LHeC, elastic EPA predictions agree with GRAPE and CepGen/LPAIR at the percent level for matched cuts and exact  $W$  kinematics.

## Elastic $\mu^+\mu^-$ benchmark at LHeC

$$Q_{e,p}^2 < 10 \text{ GeV}^2$$

|              | $W_0 = 10 \text{ GeV}$ | $W_0 = 100 \text{ GeV}$ |
|--------------|------------------------|-------------------------|
| EPA          | 120.12 pb              | 0.193 pb                |
| CepGen/LPAIR | 118.31 pb              | 0.191 pb                |
| GRAPE (BH)   | 121.37 pb              | 0.193 pb                |

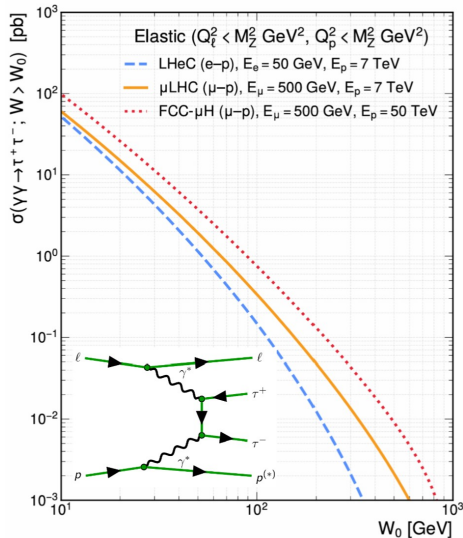
LHeC  $\gamma\gamma$  paper, Table 1 and calibration-candle discussion.



## Why it matters for LH $\mu$ C

The same QED process becomes a high-statistics calibration candle for the harder  $\mu p$  photon spectrum and any forward-tagging strategy.

# $\gamma\gamma \rightarrow \tau^+\tau^-$ : a high-statistics tau factory



Elastic cross sections from the existing HELP4CERN two-photon calculations.

## Elastic benchmark

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 50.37 \text{ pb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 58.56 \text{ pb}$$

- Large statistics make  $\tau\tau$  a natural precision channel.
- LH $\mu$ C extends the accessible  $M_{\tau\tau}$  tail.
- The hard tail enhances sensitivity to non-standard  $\gamma\tau\tau$  interactions.

# Anomalous $\tau$ electromagnetic moments: the $\gamma\tau\tau$ vertex

## General electromagnetic vertex

$$\Gamma^\mu(q^2) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2m_\tau} F_2(q^2) + \sigma^{\mu\nu}q_\nu \gamma_5 F_3(q^2),$$

$$F_2(0) = a_\tau, \quad F_3(0) = -\frac{2m_\tau}{e} d_\tau.$$

- $a_\tau$ : CP-even anomalous magnetic dipole moment.
- $d_\tau$ : CP-odd electric dipole moment.
- $\gamma\gamma \rightarrow \tau^+\tau^-$  probes the vertex directly.

Dedicated anomalous- $\tau$  study, April 2026; updated benchmark on the next slide.

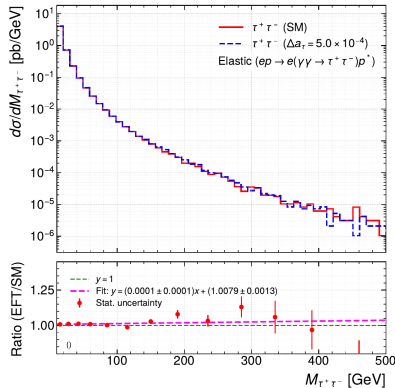
## SMEFT interpretation

Gauge-invariant tau dipoles:

$$\bar{L}_\tau \sigma^{\mu\nu} \tau_R H \left( \frac{C_{\tau B}}{\Lambda^2} B_{\mu\nu} + \frac{C_{\tau W}}{\Lambda^2} W_{\mu\nu} \right) + \text{h.c.}$$

Real and imaginary coefficient combinations map respectively to magnetic and electric dipole deformations after electroweak symmetry breaking.

# Probing $\tau$ dipole moments in the high- $M_{\tau\tau}$ tail



Updated draft\_yy\_lhec:  $\Delta a_\tau = 5 \times 10^{-4}$  and SMEFT translation.

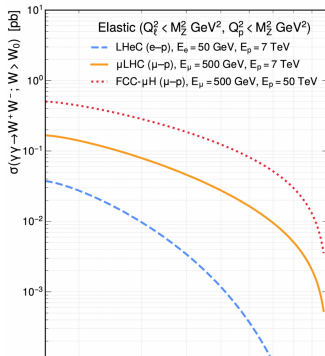
## Updated paper benchmark

The latest draft replaces the previous illustrative point by

$$\Delta a_\tau = 5.0 \times 10^{-4}.$$

- Shown: updated **LHeC** elastic reference calculation from the paper draft.
- The EFT/SM ratio exposes the growing shape distortion in the hard  $M_{\tau\tau}$  region.
- The same SMEFT fit should be repeated for  $\text{LH}\mu\text{C}$ , whose harder spectrum gives more leverage at high mass.

# $\gamma\gamma \rightarrow W^+W^-$ : a clean electroweak laboratory



Existing HELP4CERN elastic  $WW$  calculation; orange curve:  $\text{LH}\mu\text{C}@3.7$  TeV.

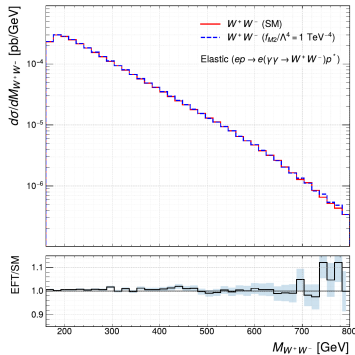
## Elastic rates

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 0.037 \text{ pb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 0.166 \text{ pb}$$

- The  $W$  threshold lies inside the high-luminosity  $\gamma\gamma$  regime.
- The process directly tests the SM  $\gamma\gamma WW$  gauge structure.
- The high- $M_{WW}$  tail is a clean lever arm for dimension-8 quartic-gauge operators.

# Anomalous quartic gauge couplings in $\gamma\gamma \rightarrow W^+W^-$



## Dimension-8 EFT

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i}{\Lambda^4} \mathcal{O}_i,$$

$$\frac{f_{M2}}{\Lambda^4} = 1 \text{ TeV}^{-4}.$$

- Shown: updated **LHeC** elastic reference from the draft.
- EFT effects are tiny at low  $M_{WW}$  and grow in the hard tail.
- $\text{LH}\mu\text{C}$  supplies the harder  $W_{\gamma\gamma}$  spectrum needed to extend this strategy quantitatively.

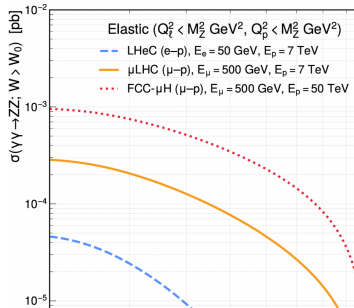
**Scope:** no detector-level  $\text{LH}\mu\text{C}$  exclusion is

claimed here; this is a defined next **HELP4CERN**

study.

Updated draft\_yy\_lhec:  $WW$  aQGC benchmark and EFT/SM ratio.

# $\gamma\gamma \rightarrow ZZ$ : a loop-induced Standard-Model channel



- In the SM,  $\gamma\gamma \rightarrow ZZ$  is absent at tree level and proceeds through loops.
- The small SM rate makes the channel a clean window on anomalous neutral quartic interactions.
- LH $\mu$ C extends the accessible high- $M_{ZZ}$  regime.

## Elastic benchmark

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 0.046 \text{ fb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 0.284 \text{ fb}$$

Existing HELP4CERN elastic ZZ calculation; updated draft motivates the neutral-QGC study.

# Neutral quartic gauge couplings in $\gamma\gamma \rightarrow ZZ$



## Rate model used in the updated study

$$\sigma_{\text{tot}} \simeq \sigma_{\text{SM}}^{\text{loop}} + \sigma_{\text{AQGC}}^{\text{tree}}.$$

## LHeC tagged-elastic reference

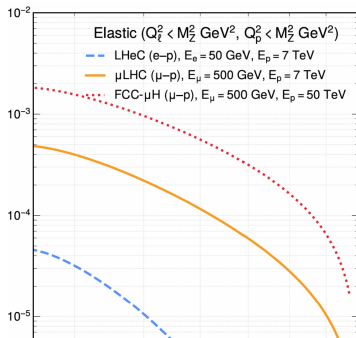
$\sigma_{\text{SM}} = 3.340 \times 10^{-5}$  pb; raw events assume  $1 \text{ ab}^{-1}$  before  $Z$  decays and detector effects.

| $f_{M2}/\Lambda^4$ | $\sigma_{\text{AQGC}}$ [pb] | events | enhancement |
|--------------------|-----------------------------|--------|-------------|
| 0.1                | $1.17 \times 10^{-8}$       | 0.012  | 1.0004      |
| 1.0                | $1.17 \times 10^{-6}$       | 1.17   | 1.035       |
| 5.0                | $2.93 \times 10^{-5}$       | 29.3   | 1.88        |

Updated draft\_yy\_lhec, neutral aQGC section.

- SM is loop induced; the quoted AQGC contribution is tree level.
- The table shows how an energy-growing neutral interaction can become competitive with a rare SM channel.
- These are **LHeC reference numbers**, not  $\text{LH}\mu\text{C}$  projected limits; the harder  $\text{LH}\mu\text{C}$  spectrum motivates the direct extension.

# Heavy final states: $\gamma\gamma \rightarrow t\bar{t}$



Existing HELP4CERN elastic  $\gamma\gamma \rightarrow t\bar{t}$  calculation.

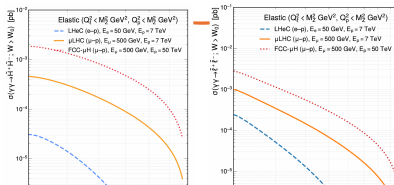
## Elastic benchmark

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 0.046 \text{ fb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 0.486 \text{ fb}$$

- This is exclusive  $\gamma\gamma \rightarrow t\bar{t}$  — **not** the much larger  $\gamma g \rightarrow t\bar{t}$  photoproduction channel.
- The  $2m_t$  threshold probes the hard end of the photon spectrum.
- The channel accesses the top electromagnetic interaction and high-energy new physics.

# Beyond the Standard Model: charged-particle pair production in $\gamma\gamma$ collisions



Illustrative

100 GeV scalar (slepton) and fermion (higgsino) reference curves.

- Representative channels:

$$\gamma\gamma \rightarrow \tilde{\ell}^+ \tilde{\ell}^-, \quad \gamma\gamma \rightarrow \tilde{\chi}^+ \tilde{\chi}^-.$$

- Rates are governed mainly by electric charge and mass, complementary to strongly interacting searches.
- Forward kinematic closure is attractive for compressed spectra with soft visible products and missing momentum.

## Updated interpretation

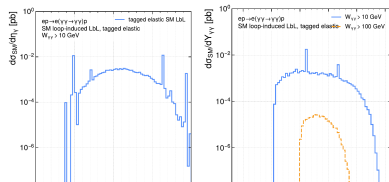
The draft notes that 100 GeV is now illustrative. Current constraints motivate roughly 300 GeV sleptons and 250–300 GeV higgsino benchmarks; LHeC rates near those thresholds are tiny, making a dedicated LH $\mu$ C re-evaluation especially relevant.

yy\_lhec SUSY section + updated compressed-SUSY discussion in draft\_yy\_lhec.

# Light-by-light scattering, Higgs production, and other $\gamma\gamma$ opportunities

## Mature calculation: $\gamma\gamma \rightarrow \gamma\gamma$

SM light-by-light scattering is loop induced; all charged SM particles contribute to the box amplitude.



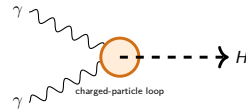
Updated

LHeC tagged-elastic SM pseudorapidity distributions.

Updated draft\_yy\_lhec LbL section; Higgs-via- $\gamma\gamma$  section in yy\_lhec / draft\_yy\_lhec.

## Exploratory: $\gamma\gamma \rightarrow H$

The Higgs channel is loop induced and rare, but it probes the effective  $H\gamma\gamma$  interaction in an exceptionally clean initial state.



## LH $\mu$ C opportunity

The harder photon spectrum extends the high-mass LbL region and other loop-induced or resonant searches.

# What does two-photon physics require from the $\text{LH}\mu\text{C}$ experiment?



- **Wide central tracking and lepton acceptance**, including robust  $\tau$  reconstruction.
- **Efficient  $W/Z/t$  reconstruction** for heavy  $\gamma\gamma$  final states.
- **Exclusivity and acoplanarity** observables with sensitivity to low extra activity.
- **Missing-momentum performance** for neutrinos and compressed BSM signatures.
- **Scattered-muon measurement** to constrain the lepton-side photon kinematics.
- **Forward proton tagging where feasible**, together with control of proton dissociation.
- Exploit the defining advantage: the **clean, low-pile-up  $\mu p$  environment**.

## Figure placeholder

Suggested figure:  $\text{LH}\mu\text{C}$  interaction-region schematic with central detector, scattered-muon acceptance and a near-beam proton tag.

## Physics–detector connection

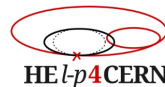
The  $\gamma\gamma$  programme links central reconstruction, lepton tagging and near-beam instrumentation directly to the usable photon luminosity and kinematic closure.

# Summary and outlook



- $\text{LH}\mu\text{C}$  turns  $\mu p$  collisions into a **high-energy  $\gamma\gamma$  laboratory** with a substantially harder photon spectrum than the LHeC benchmark.
- EPA and  $S_{\gamma\gamma}$  connect parent  $\mu p$  luminosity transparently to elastic and proton-dissociative two-photon subprocesses.
- $\tau^+\tau^-$  and  $W^+W^-$  provide high-statistics precision/EFT channels; sensitivity lives in their hard invariant-mass tails.
- Loop-induced  $ZZ$ , exclusive  $t\bar{t}$ , light-by-light scattering,  $H$ , and charged BSM pairs extend the programme to rare quantum processes and heavy thresholds.

**Bottom line: high-energy  $\gamma\gamma$  physics is one of the distinctive physics opportunities of  $\text{LH}\mu\text{C}$ .**



## Thank you!

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